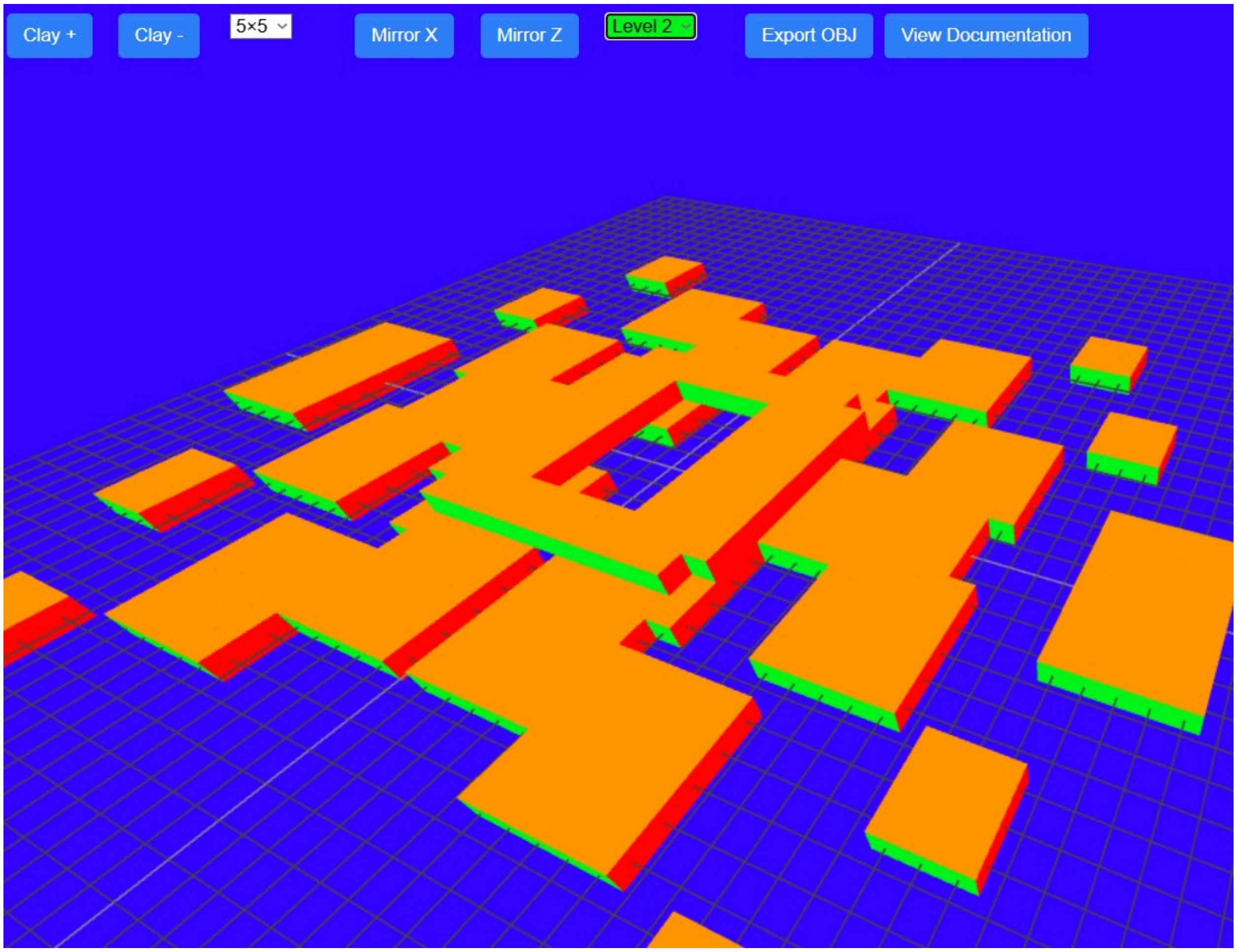

VisionCraft 3D

An Augmentation & Accessibility Prototype by Joshua Bohumil

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3D Digital Building Hobby for People with Colorblindness

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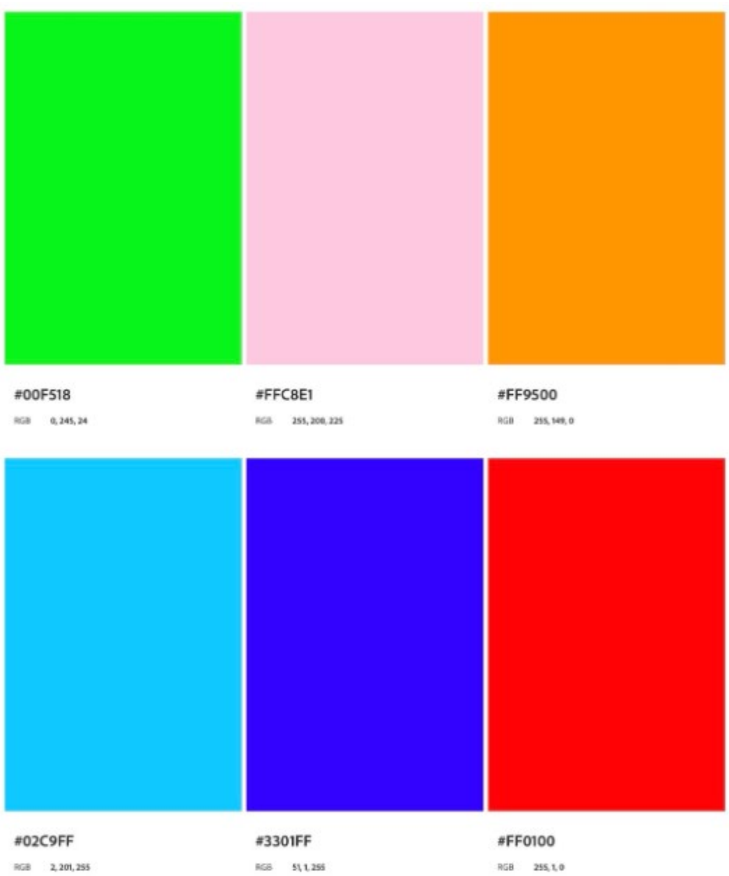
The Augmentation

VisionCraft 3D is a web-based sculpting tool designed to make creating digital 3D art out of cubes more accessible for people with incomplete achromatopsia. Incomplete achromatopsia is a rare type of colorblindness where it is difficult to tell a difference in colors when the colors do not have a lot of contrast. Developing for people with this condition limits the color palette that should be incorporated in the design.

Building VisionCraft 3D

I used Adobe Color to find a color palette that would work. I vibe-coded HTML code for the web-based tool. Vibe-coding is a way of developing websites by guiding an AI through typed prompts. Then each AI generated code was tested. I thought critically about the results and edited the code to create a final product. By building a working prototype this way, I learned a lot about HTML code, AI, and incomplete achromatopsia. The image below shows the color palette I made, and Adobe Color determined it was colorblind safe.



Initially the idea was to use this color palette differently. Each cube on the first level was orange, each cube on the second level green, and so on. I realized that people with this particular type of colorblindness would probably have a difficult time seeing shadows of the same color. This led me to create a new version where pairs of opposing sides were the same color.

Augmentation Technologies Defined

Augmentation technologies are tools, either physical or digital, that help people think, feel, move, or sense things in new or improved ways. When people hear about these types of technologies for the first time they may be concerned about privacy and dependence. Companies should not collect unnecessary information about their users. Users also should not become too dependent on current technology as technology is always changing.

The Experience

Playing with blocks as a child has been found to be beneficial for development. It has also been found that video games such as Minecraft share the same beneficial experiences. The use of blocks, even in a digital environment, teaches spatial reasoning and a basic understanding of geometry [1].

VisionCraft 3D gives children and adults this experience in a simple to use website. VisionCraft 3D does not limit the user just creating digital 3D art, they can also export an OBJ file and 3D print their creation. For adult creators they can use this export of their creations to then sell them on 3D asset marketplaces or potentially animate their artwork using other software.

VisionCraft 3D allows the user to build on 8 levels vertically. The cubes snap to a grid. That makes building easier. The cubes are referred to as clay. This is a reference to traditional clay sculpture. Clay can be added or subtracted from other clay. The user simply taps or clicks on where in the grid they want their cubes.

Accessibility, Usability, and User Experience

VisionCraft 3D can be accessed by anyone with internet access and a web browser. This means users can bring up the tool on their tablets, smart phones, and computers. VisionCraft 3D has built in documentation. Users can click the "View Documentation" button to understand how to use this tool. The tool was designed with touchscreens as well as mouse and keyboards in mind. I personally found touchscreens to be easier to work with. However, by putting the word "recommended" next to "Camera controls On Touchscreen" I inadvertently may have discouraged users from using mouse and keyboard. Everyone is different and some users may find the mouse and keyboard controls easier than using a touchscreen.

This tool has the potential to evolve into a collaborative experience where friends can build together on the same project. This would also help develop communication and teamwork skills. In the future I hope to create software similar to this one that allows users with this condition to 3D print, sell, and animate their designs. This would open up new possibilities and hobbies for users of VisionCraft 3D.

Tradeoffs

While users can export their creations for a number of purposes, they still may require assistance to use other software for 3D printing, selling, and animating. This means some users will feel like an interesting featuring is being locked for them due to needing to ask for support. I wanted to add a feature that allowed users to upload their drawings to a transparent plane at the center of the grid. These photos would stand vertically along the x and y axes. This would help users make realistic objects like architecture, animals, and characters. There were multiple reasons I did not add this feature. I was not sure the photos would scale in the tool, and I ran out of time for this version of the prototype.

Why It Matters

VisionCraft 3D is an overall ethical tool, that was co-created with AI and implements accessibility. Tools like this one should be built with privacy for the user in mind. This tool does not sell the users information. It does not ask security questions for any user. There is no sign up or login. It does not spy on the user or recommend the next clay cube to place and where. This is an example of non-intrusive design which is beneficial and ethical for the user. The tool is predictable, visible, and understandable, with no hidden processes. However, people should not rely too much on AI when developing tools like this. This is because the developer needs to have at least a basic understanding of how the site and code works if something were to go wrong. I hope that in the future we are not all scratching our heads wondering what AI has built for us and taking it for granted. It is up to humans to tell AI what makes a site ethical. That being said, AI is a tool that can be used to increase one's knowledge of how code works. It is not always perfect, but I feel like I have learned the basics of HTML by talking with the AI and having moments where I had to step in and fix the code myself. AI has the potential to create many tools like this and help many people with their needs.

VisionCraft 3D QR Code & Link

https://wonderful-sawine-dbc713.netlify.app/

References

[1]

https://attunedpsychology.com/building-blocks-minecraft-may-actually-helping-child-learn/